Maryn O. Carlson

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EDUCATION

University of Chicago *Fall 2017 - Summer 2023*
PhD in Genetics, Genomics, and Systems Biology
Thesis: "Two problems in population genetics."
Advisor: Matthias Steinrücken

Cornell University *Fall 2013 - Summer 2016*
MSc in Plant Pathology and Plant-Microbe Biology
Thesis: "Temporal Genetic Dynamics and Mating Type Inheritance in an Experimental Biparental Field Population of *Phytophthora capsici*."
Advisors: Christine Smart & Michael Gore

Cornell University *Winter 2011 - Winter 2013*
BS, Interdisciplinary Studies, *cum laude*, four semesters

Columbia University *Fall 2009 - Spring 2010*
Freshman Year

PUBLICATIONS

1. Brooks C, Desai MM, **Carlson MO**. (*In draft*). Mutation choice biases the structure of empirical fitness landscapes.
2. **Carlson MO**, Andrews B, and Simons YB. (2025). Distinguishing direct interactions from global epistasis using rank statistics. *Proceedings of the National Academy of Sciences*. 122(39):e2509444122.
3. **Carlson MO**. (2023). Two problems in population genetics. *PhD Thesis*. The University of Chicago.
4. **Carlson MO**, Rice DP, Berg JJ, and Steinrücken M. (2022). Polygenic score accuracy in ancient samples: Quantifying the effects of allelic turnover. *PLOS Genetics*. 18(5):e1010170.
5. Stajich JE, Vu AL, Judelson HS, Vogel GM, Gore MA, **Carlson MO**, Devitt N, Jacobi J, Mudge J, Lamour KH, Smart CD. (2021). High quality reference genome sequence for the oomycete vegetable pathogen *Phytophthora capsici* strain LT1534. *Microbiology Resource Announcements*. 10(21):e00295-21.
6. **Carlson MO**, Montilla-Bascon G, Hoekenga OA, Tinker NA, Poland J, Baseggio M, Sorrells ME, Jannink JL, Gore MA and Yeats TH. (2019). Multivariate genome-wide association analyses reveal the genetic basis of seed fatty acid composition in oat (*Avena sativa* l.). *G3: Genes, Genomes, Genetics*. 9(9):2963-2975.
7. **Carlson MO**, Gazave E, Gore MA, and Smart CD. (2017). Temporal genetic dynamics of an experimental, biparental field population of *Phytophthora capsici*. *Frontiers in Genetics*. 8:10.3389/fgene.2017.00026.

8. **Carlson MO**. (2016). Temporal genetic dynamics of an experimental, biparental field population of *Phytophthora capsici*. *Master's thesis.* Cornell University.

9. Lange HW, Tancos MA, **Carlson MO**, and Smart CD. (2016). Diversity of *Xanthomonas campestris* isolates from symptomatic crucifers in New York State. *Phytopathology.* 106:113-122.

10. Summers CF, Gulliford CM, Carlson CH, Lillis JA, **Carlson MO**, Cadle-Davidson L., Gent DH, and Smart CD. (2015). Identification of genetic variation between obligate plant pathogens *Pseudoperonospora cubensis* and *P. humuli* using RNA sequencing and genotyping-by-sequencing. *PLOS One.* 10(11):e0143665.

HONORS & AWARDS

Postdoctoral Fellowship, NSF-Simons National Institute for Theory and Mathematics in Biology	2024
Harper Award Finalist, University of Chicago	2022
Young Investigator Travel Award, Society of Molecular Biology and Evolution	2020
2018-2019 Dean's Outstanding Teaching Assistant Prize	2019
• Awarded annually to two graduate students in the Biological Sciences Division (*Description*).	
Centennial Fellowship, Princeton University (Declined)	2017
Travel Award, International Society Molecular Plant-Microbe Interactions	2016
Honorable Mention, National Science Foundation GRFP	2014
$10,000 Annie's Sustainable Agriculture Scholarship	2012

RESEARCH EXPERIENCE

Postdoctoral fellow, NSF-Simons NITMB June 2024 - present
Independent fellow

- Developed a semi-parametric method for detecting direct interactions between mutations in the presence of global epistasis from combinatorial experiments.
- Developing non- and semi-parametric methods for characterizing genotype-to-phenotype maps.
- Using random fitness models to quantify how adaptation distorts our "picture" of the underlying fitness landscape and inference of epistatic interactions among mutations.

Postdoctoral scholar, University of Chicago July 2023 - June 2024
Lab of Arvind Murugan

- Studied complex network design and evolution in order to understand the emergence of many-to-many biological signaling networks.

Graduate researcher, University of Chicago October 2018 - July 2023
Lab of Matthias Steinrücken

- Designed and applied a novel statistical inference procedure to identify loss of heterozygosity events in, and the ancestral state of, clonal lineages of the plant pathogen *Phytophthora capsici*.
- Quantified polygenic score accuracy in out-of-sample predictions in a theoretical framework.

Research scientist, Cornell University August 2016 - August 2017
Labs of Michael Gore, Jean-Luc Jannink, and Mark Sorrells

- Utilized high-dimensional metabolite data and GWAS to identify loci associated with metabolite variation in cultivated oat, while controlling for batch effects in the metabolite data due to sample processing.

Graduate research assistant, Cornell University August 2013 - August 2016
Labs of Michael Gore and Christine Smart

- Analyzed time-series genetic data from an experimental plant pathogen population, identifying putatively selected loci and localizing the mating type determining region.
- Identified regions that likely underwent mitotic loss of heterozygosity.

Lab technician, Cornell University January 2013 - July 2013
Lab of Harold van Es

- Evaluated the accuracy of software that provides optimal fertilizer recommendations to corn growers.

TEACHING ASSISTANTSHIPS

Population Genetics, University of Chicago *Winters 2020, 2021*
Fundamentals of Computational Biology: Models and Inference, U. of Chicago *Winter 2019*
Quantitative Biology (QBio) Bootcamp, University of Chicago *September 2018*
Biology and Management of Plant Diseases, Cornell University *Fall 2014*

ADDITIONAL TEACHING & MENTORING EXPERIENCE

Tutor, *Intro to Pop. Gen. & Fundamentals of Comp. Bio.*, University of Chicago *Winter 2023*
Tutor, *Numerical Linear Algebra*, University of Chicago *Fall 2018*
Teaching Assistant, Software Carpentry Workshops, University of Chicago *2017 & 2018*
Instructor, *Introduction to the R Programming Language*, Cornell University *August 2015*

- Designed and taught a four-part introductory programming course for $\approx$ 25 undergraduate, graduate students, and staff.

Mentor for undergraduate students, Cornell University *2014-2015*

- Supervised two students during successive summer undergraduate research programs.

Mentor for high school students, Cornell University *2014-2015*

- Helped to supervise two students from Geneva High School.

INVITED & EXTERNAL PRESENTATIONS

1. "When are two mutations epistatic?" Department of Biology Seminar, University of Rochester. (Invited talk), February 23, 2026.
2. "When are two mutations epistatic?" Institute for Human Genetics Seminar, Clemson University. (Invited talk), January 20, 2026.
3. "When are two mutations epistatic?" Department of Biology Seminar, University of Texas at Arlington. (Invited talk), December 8, 2025.
4. "Robust detection of specific epistasis." Probabilistic Modeling in Genomics, Cold Spring Harbor Laboratory. (Poster), 2025.
5. "Loss of heterozygosity in a plant pathogen: Methods & inference." Probabilistic Modeling in Genomics, Cold Spring Harbor Laboratory. (Poster), 2023.
6. "Polygenic scores through time: Quantifying the contribution of allelic turnover." Quantitative Biology Bootcamp 8, University of Chicago. (Invited talk), 2022.

7. "Polygenic scores through time: Quantifying the contribution of allelic turnover." Probabilistic Modeling in Genomics, Cold Spring Harbor Laboratory. (Virtual poster), 2021.
8. "Polygenic scores through time: quantifying the contribution of allelic turnover." Annual Meeting of the Society of Molecular Biology and Evolution, Quebec City, Canada. (Oral), 2020–canceled.
9. "Polygenic scores in time: Quantifying the contribution of allelic turnover to prediction bias." Midwest Population Genetics VI. (Poster), 2019.
10. "Unraveling fatty acid variation in oat (*Avena sativa* L.) with multivariate genome-wide association analyses." Population Evolutionary and Quantitative Genetics Conference. (Poster), 2018 (315T).
11. "Genetic dynamics and mating type inheritance in an experimental overwintering biparental population of *Phytophthora capsici* in the context of a field infestation." Department Seminar Series. Pennsylvania State University. (Invited talk), 2016.
12. "Genetic dynamics and mating type inheritance in an experimental overwintering biparental population of *Phytophthora capsici*." Molecular Plant Microbe Interactions (MPMI) Congress, Portland, OR. (Poster), 2016 (P13-410).
13. "Genetic dynamics of an experimental overwintering biparental population in the context of a field infestation of *Phytophthora capsici*." Soilborne Oomycete Conference. Duck Key, FL. (Talk), 2015.
14. "Population genomics of an overwintering biparental *Phytophthora capsici* population." American Phytopathological Society Conference. Pasadena, CA. (Talk), 2015 (18-0).

* I have also given many internal talks at NITMB, University of Chicago, and Cornell University.

OUTREACH & SERVICE

Reviewed manuscripts for *Genetics*, *Plos Comp. Bio.*, *Comm. Biology*, *Sci. Reports*, *Plant Disease* — *2016-present*
Organized an informal, weekly "theory" reading group at University of Chicago — *2025-2026*
Organized & ran a weekly *Research-in-Progress* seminar series at NITMB, Chicago, IL — *2024-2025*
Helped start the Center For Living Systems bi-weekly chalk talk series at University of Chicago — *2024*